

fastp report

Summary

General

fastp version:	1.0.1 (https://github.com/OpenGene/fastp)
sequencing:	paired end (294 cycles + 294 cycles)
mean length before filtering:	263bp, 293bp
mean length after filtering:	259bp, 259bp
duplication rate:	0.006667%
Insert size peak:	434
Detected read2 adapter:	GATCGGAAGAGCGTCGTGTAGGGAAAGAGTGT ->Nextera_LMP_Read2_External_Adapter

Before filtering

total reads:	30.000000 K
total bases:	8.355356 M
Q20 bases:	7.551286 M (90.376592%)
Q30 bases:	6.801583 M (81.403868%)
Q40 bases:	0 (0.000000%)
GC content:	38.639658%

After filtering

total reads:	27.790000 K
total bases:	7.209557 M
Q20 bases:	6.841759 M (94.898466%)
Q30 bases:	6.263531 M (86.878167%)
Q40 bases:	0 (0.000000%)
GC content:	38.221294%

Filtering result

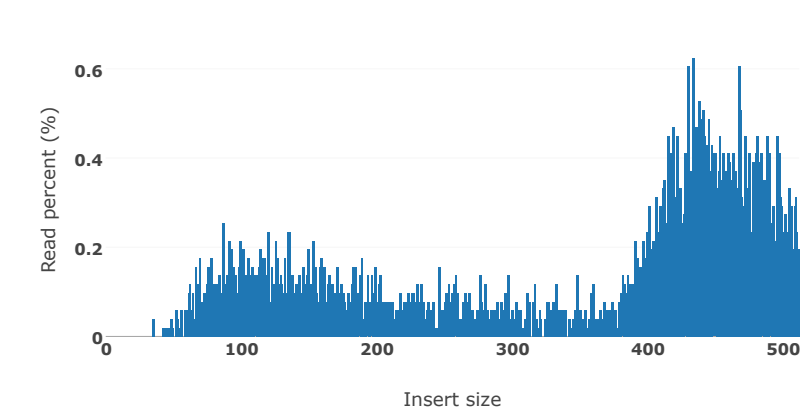
reads passed filters:	27.790000 K (92.633333%)
reads with low quality:	2.142000 K (7.140000%)
reads with too many N:	6 (0.020000%)
reads too short:	62 (0.206667%)

Adapters

Adapter or bad ligation of read1 The input has little adapter percentage (~0.857821%), probably it's trimmed before.		Adapter or bad ligation of read2	
Sequence	Occurrences	Sequence	Occurrences
ACCTTAG	3245	ACCTTAG	78
other adapter sequences	268	other adapter sequences	3457

Insert size estimation

Insert size distribution (32.421875% reads are with unknown length)

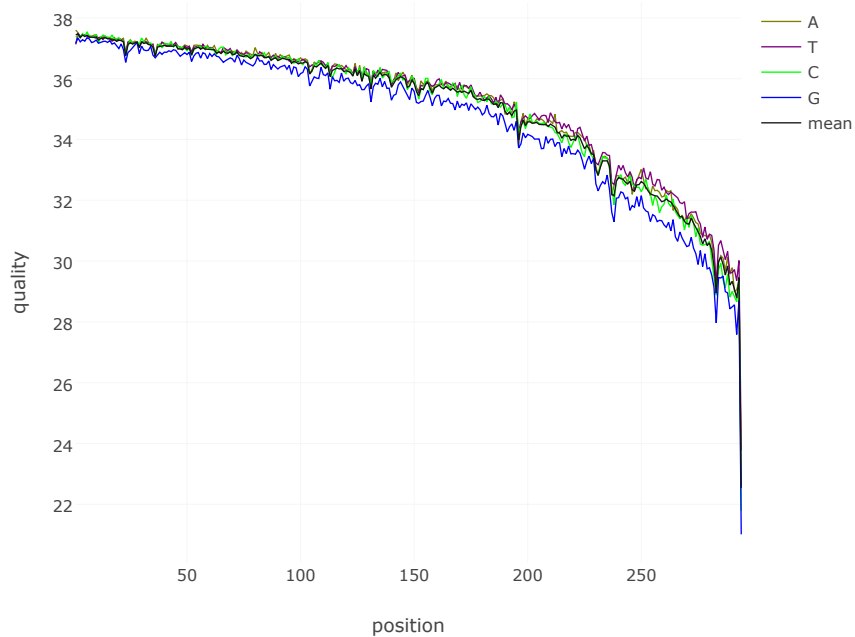


This estimation is based on paired-end overlap analysis, and there are 32.421875% reads found not overlapped. The nonoverlapped read pairs may have insert size <30 or >558, or contain too much sequencing errors to be detected as overlapped.

Filtering statistics

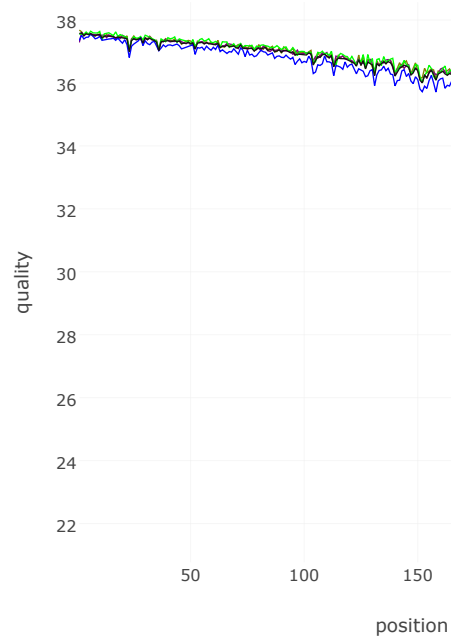
Before filtering: read1: quality

Value of each position will be shown on mouse over.



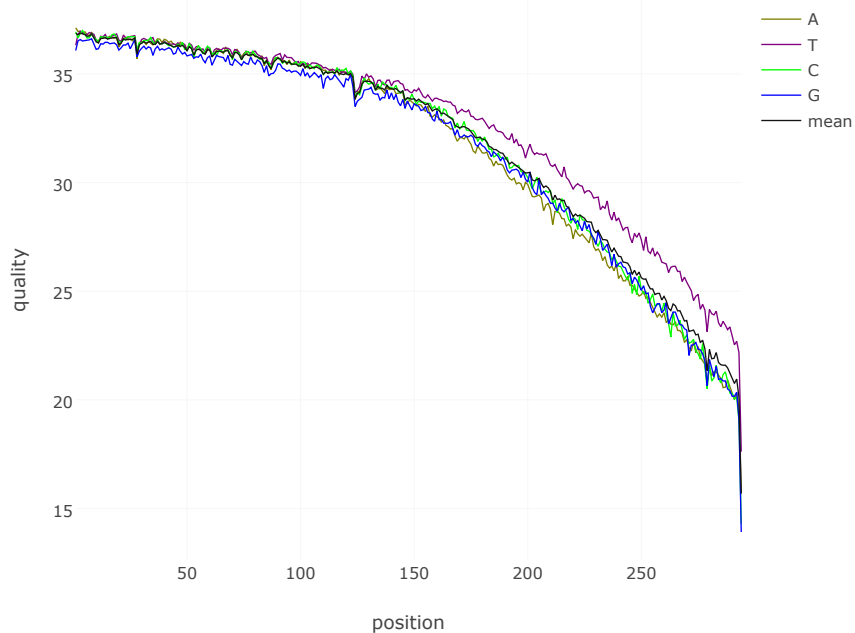
After filtering: read1: quality

Value of each position will be shown on mouse over.



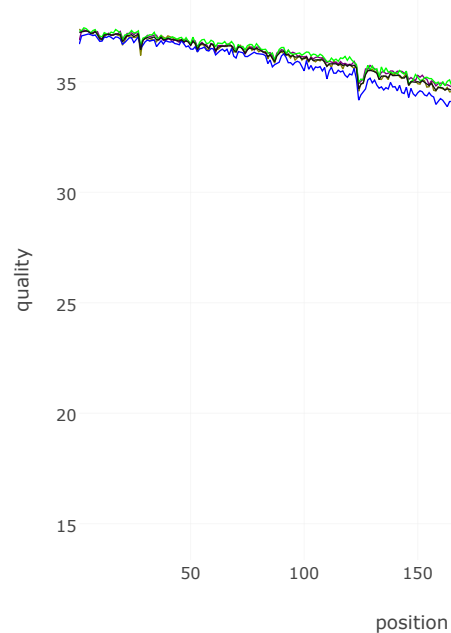
Before filtering: read2: quality

Value of each position will be shown on mouse over.

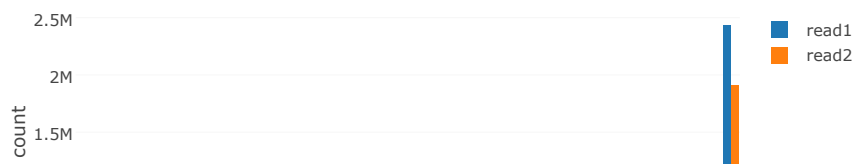


After filtering: read2: quality

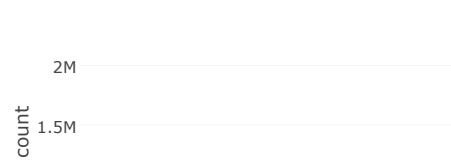
Value of each position will be shown on mouse over.

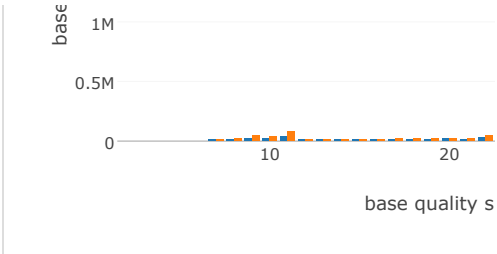
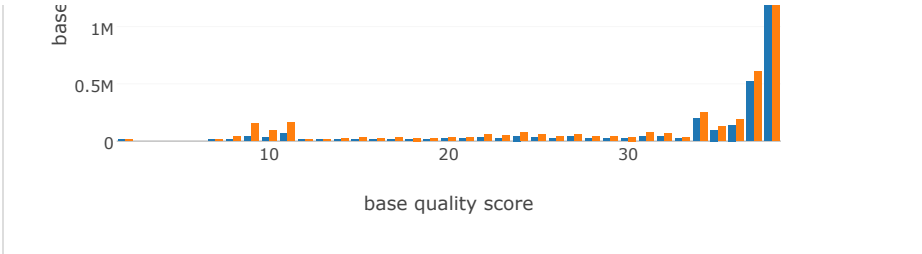


Before filtering: quality score histogram



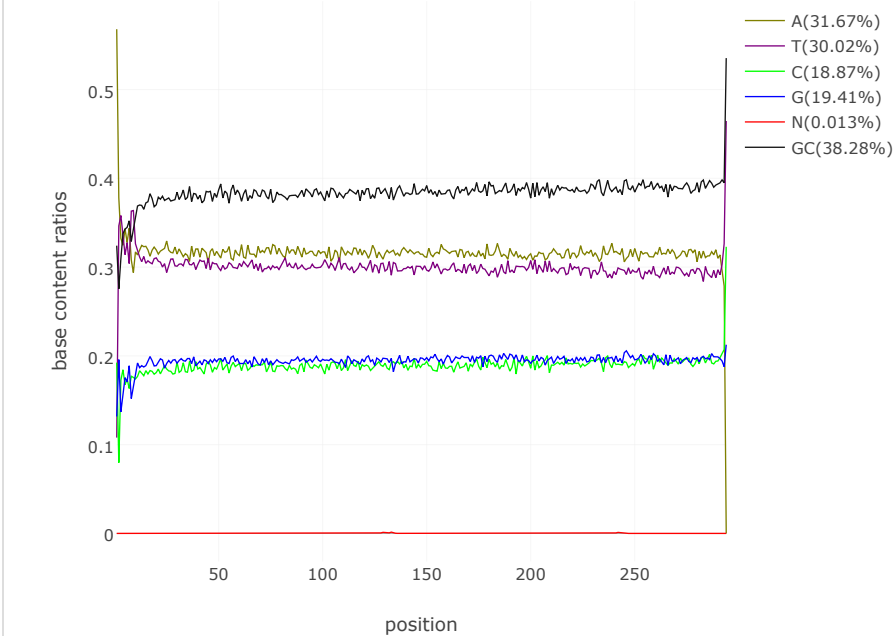
After filtering: quality





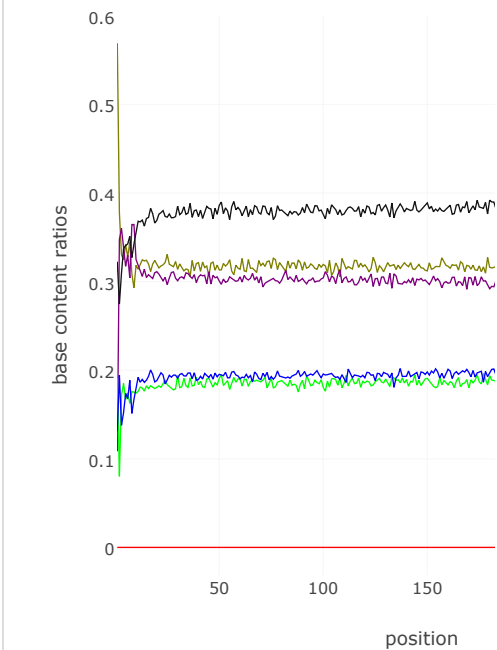
Before filtering: read1: base contents

Value of each position will be shown on mouse over.



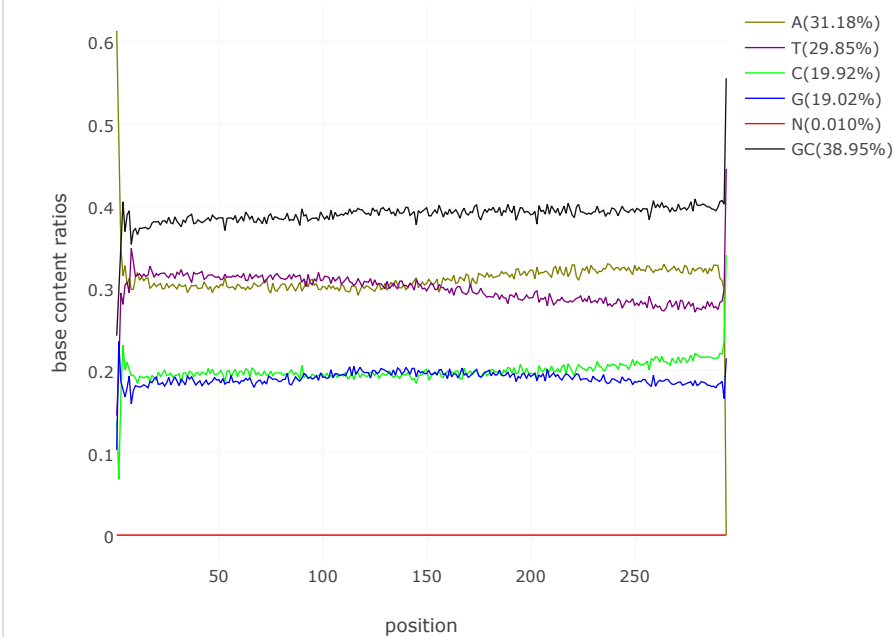
After filtering: read1: base contents

Value of each position will be shown on mouse over.



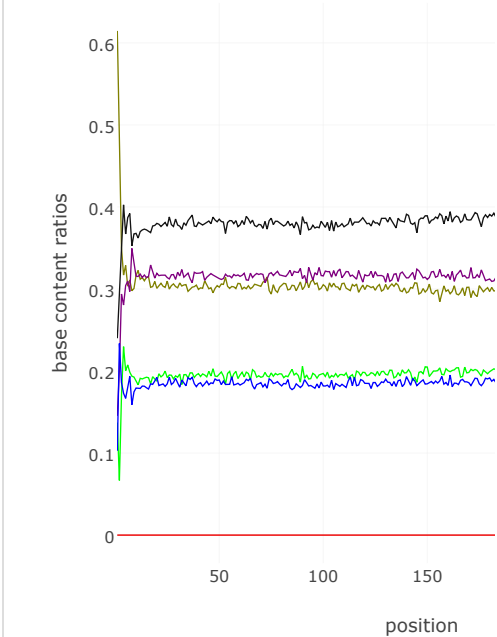
Before filtering: read2: base contents

Value of each position will be shown on mouse over.



After filtering: read2: base contents

Value of each position will be shown on mouse over.



After filtering: read1: KMER counting

Darker background means larger counts. The count will be shown

	AA	AT	AC	AG	TA	TT	TC	TAAT
AAA	AAAA	AAAT	AAAC	AAAG	AAATA	AAATT	AAATC	AAATG
AAT	ATAAT	ATAAT	ATAAC	AATAG	AATTA	AATTT	AATTC	AATTG
AA	AAAT	AAAT	AAAT	AAAG	AAATA	AAATT	AAATC	AAATG
AGA	AGAAA	AGAAAT	AGAAC	AGAG	AGAGT	AGAGT	AGAGT	AGAGT
ATA	ATAAA	ATAAT	ATAAC	ATAAG	ATATA	ATATT	ATATC	ATATG
ATT	ATTAA	ATTAT	ATTAC	ATTAG	ATTTA	ATTTT	ATTTC	ATTTG
ATC	ATCAA	ATCAT	ATCAC	ATCAG	ATCTA	ATCTT	ATCTC	ATCTG
ATG	ATGAA	ATGAT	ATGAC	ATGAG	ATGTA	ATGTT	ATGTC	ATGTG
ACAA	ACAAA	ACAAT	ACAC	ACAG	ACATA	ACATT	ACATC	ACATG
ACC	ACCTA	ACCAT	ACCTAC	ACCTAG	ACCTTA	ACCTTT	ACCTTC	ACCTTG
ACC	ACCAA	ACCAAT	ACCAC	ACCAG	ACCGA	ACCGT	ACCGT	ACCGT
ACG	ACGAA	ACGAT	ACGAC	ACGAG	ACGTA	ACGTT	ACGTC	ACGTG
AGA	AGAAA	AGAAAT	AGAAC	AGAG	AGAGT	AGAGT	AGAGT	AGAGT
AGT	AGTAA	AGTAT	AGTAC	AGTAG	AGTTA	AGTTT	AGTTC	AGTTG
AG	AGAT	AGAT	AGAT	AGAG	AGATA	AGATT	AGATC	AGATG
AGG	AGGAA	AGGAAT	AGGAC	AGGAG	AGGAGT	AGGAGT	AGGAGT	AGGAGT
TAA	TAAAA	TAAAT	TAAAC	TAAAG	TAAAT	TAAAT	TAAAT	TAAAT
TAT	TATAA	TATAT	TATAC	TATAG	TATTA	TATTT	TATTC	TATTG
TAC	TACAA	TACAT	TACAC	TACAG	TACTA	TACTT	TACTC	TACTG
TAG	TAGAA	TAGAT	TAGAC	TAGAG	TAGTA	TAGTT	TAGTC	TAGTG
TAT	TATAA	TATAT	TATAC	TATAG	TATTA	TATTT	TATTC	TATTG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTTA	TTTTT	TTTTTC	TTTTTG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTTA	TTTTT	TTTTTC	TTTTTG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTTA	TTTTT	TTTTTC	TTTTTG
TTG	TTGAA	TTGAT	TTGAC	TTGAG	TTGTA	TTGTT	TTGTC	TTGTG
TCA	TCAAA	TCAAT	TCAAC	TCAG	TCATA	TCATT	TCATC	TCATG
TC	TCAT	TCAT	TCAT	TCAG	TCATA	TCATT	TCATC	TCATG
TCG	TCGAA	TCGAT	TCGAC	TCGAG	TCGTA	TCGTT	TCGTC	TCGTG
TCG	TCGAA	TCGAT	TCGAC	TCGAG	TCGTA	TCGTT	TCGTC	TCGTG
TGA	TGAAA	TGAAT	TGAAC	TGAG	TGATA	TGATT	TGATC	TGATG
TGT	TGTAA	TGTAT	TGTAC	TGTAG	TGTTA	TGTTT	TGTTC	TGTTG
TG	TGCA	TGCA	TGCA	TGCA	TGCTA	TGCTT	TGCTC	TGCTG
TGG	TGGAA	TGGAT	TGGAC	TGGAG	TGGTA	TGGTT	TGGTC	TGGTG
C	CAAAA	CAAAAT	CAAAAC	CAAAAG	CAAAATA	CAAAATT	CAAAATC	CAAAATG
CAT	CATAA	CATAT	CATAC	CATAG	CATTA	CATTT	CATTC	CATTG
CAC	CACAA	CACAT	CACAC	CACAG	CACATA	CACATC	CACATC	CACATC
CAG	CAGAA	CAGAT	CAGAC	CAGAG	CAGTA	CAGTT	CAGTC	CAGTG
CTA	CTAAA	CTAAT	CTAAC	CTAG	CTATA	CTATT	CTATC	CTATG
CTT	CTTAA	CTTAT	CTTAC	CTTAG	CTTTA	CTTTT	CTTTTC	CTTTTG
CTC	CTCAA	CTCAT	CTCAC	CTCAG	CTCTA	CTCTT	CTCTC	CTCTG
CTG	CTGAA	CTGAT	CTGAC	CTGAG	CTGTA	CTGTT	CTGTC	CTGTG
C	CAAAA	CAAAAT	CAAAAC	CAAAAG	CAAAATA	CAAAATT	CAAAATC	CAAAATG
CCA	CCATA	CCAT	CCAT	CCAT	CCAT	CCAT	CCAT	CCAT
CCG	CCGAA	CCGAT	CCGAC	CCGAG	CCGTA	CCGTT	CCGTC	CCGTG
CCG	CCGAA	CCGAT	CCGAC	CCGAG	CCGTA	CCGTT	CCGTC	CCGTG
CGT	CGTAA	CGTAT	CGTAC	CGTAG	CGTTA	CGTTT	CGTTC	CGTTG
CGC	CGCAA	CGCAT	CGCAC	CGCAG	CGCTA	CGCTT	CGCTC	CGCTG
CGG	CGGAA	CGGAT	CGGAC	CGGAG	CGGTA	CGGTT	CGGTC	CGGTG
GAA	GA AAA	GAAT	GAAC	GAAG	GAATA	GAATT	GAATC	GAATG
GAT	GATAA	GATAT	GATAC	GATAG	GATTA	GATTT	GATTC	GATTG
GAC	GACAA	GACAT	GACAC	GACAG	GACATA	GACATC	GACATC	GACATC
GAG	GAGAA	GAGAT	GAGAC	GAGAG	GAGTA	GAGTT	GAGTC	GAGTG
GTA	GTA AA	GTAAT	GTAAC	GTAAG	GTAATA	GTAATT	GTAATC	GTAATG
GTT	GT TAA	GT TAT	GT TAC	GT TAG	GT TTA	GT TTT	GT TTT	GT TTT
GTC	GTC AA	GTCAT	GTCAC	GTCAG	GTCATA	GTCATT	GTCATC	GTCATG
GTT	GTTAA	GTTAT	GTTAC	GTTAG	GTTTA	GTTTT	GTTTC	GTTTG
GCA	GCA AA	GCAAT	GCAAC	GCAAG	GCAATA	GCAATT	GCAATC	GCAATG
GCT	GCT AA	GCTAT	GCTAC	GCTAG	G			

After filtering: read2: KMER counting

Darker background means larger counts. The count will be shown

AAA	AA	AT	AC	AG	TA	TT	TC	TG
AAA	AAAA	AAAT	AAAC	AAAG	AAATA	AAATT	AAATC	AAATG
AAT	AATAA	AATAT	AATAC	AATAG	AATATA	AATATT	AATATC	AATATG
ACA	ACAAA	ACAAT	ACAC	ACAG	ACATA	ACATT	ACATC	ACATG
AGA	AGAAA	AGAA	AGAC	AGAG	AGATA	AGATT	AGATC	AGATG
ATA	ATAAA	ATAAT	ATAAC	ATAAG	ATATA	ATATT	ATATC	ATATG
ATT	ATTAA	ATTAT	ATTAC	ATTAG	ATTATA	ATTATT	ATTATC	ATTATG
ATC	ATCAA	ATCAT	ATCAC	ATCAG	ATCTA	ATCTT	ATCTC	ATCTG
ATG	ATGAA	ATGAT	ATGAC	ATGAG	ATGATA	ATGATT	ATGATC	ATGATG
ACC	ACCAA	ACCAT	ACCA	ACAG	ACATA	ACATT	ACATC	ACATG
ACT	ACTAA	ACTAT	ACTAC	ACTAG	ACTATA	ACTATT	ACTATC	ACTATG
ACC	ACCAA	ACCAT	ACCAC	ACCAG	ACCTA	ACCTT	ACCTC	ACCTG
ACG	ACGAA	ACGAT	ACGAC	ACGAG	ACGATA	ACGATT	ACGATC	ACGATG
AGA	AGAAA	AGAA	AGAC	AGAG	AGATA	AGATT	AGATC	AGATG
AGC	AGCAA	AGCAT	AGCAC	AGCAG	AGCTA	AGCTT	AGCTC	AGCTG
AGG	AGGAA	AGGAT	AGGAC	AGGAG	AGGATA	AGGATT	AGGATC	AGGATG
TAA	TAAAA	TAAAT	TAAAC	TAAAG	TAAATA	TAAATT	TAAATC	TAAATG
TAT	TATAA	TATAT	TATAC	TATAG	TATATA	TATATT	TATATC	TATATG
TAC	TACAA	TACAT	TACAC	TACAG	TACTA	TACTT	TACTC	TACTG
TAG	TAGAA	TAGAT	TAGAC	TAGAG	TAGATA	TAGATT	TAGATC	TAGATG
TAA	TAAAA	TAAAT	TAAAC	TAAAG	TAAATA	TAAATT	TAAATC	TAAATG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTATA	TTTATT	TTTATC	TTTATG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTATA	TTTATT	TTTATC	TTTATG
TTT	TTTAA	TTTAT	TTTAC	TTTAG	TTTATA	TTTATT	TTTATC	TTTATG
TGA	TGAAA	TGAAT	TGAAC	TGAAG	TGATA	TGATT	TGATC	TGATG
TGT	TGTAA	TGTAT	TGTAC	TGTAG	TGTATA	TGTATT	TGTATC	TGTATG
TGC	TGCAA	TGCAT	TGCAC	TGCAG	TGCTA	TGCTT	TGCTC	TGCTG
TCA	TCAAA	TCAAT	TCAAC	TCAAG	TCAT	TCAAT	TCACT	TCAAG
TCT	TCTAA	TCTAT	TCTAC	TCTAG	TCTATA	TCTATT	TCTATC	TCTATG
TCC	TCCAA	TCCAT	TCCAC	TCCAG	TCCTA	TCCTT	TCCTC	TCCTG
TGC	TGCAA	TGCAT	TGCAC	TGCAG	TGCT	TGCTT	TGCTC	TGCTG
TGA	TGAAA	TGAAT	TGAAC	TGAAG	TGATA	TGATT	TGATC	TGATG
TGT	TGTAA	TGTAT	TGTAC	TGTAG	TGTATA	TGTATT	TGTATC	TGTATG
TGC	TGCAA	TGCAT	TGCAC	TGCAG	TGCT	TGCTT	TGCTC	TGCTG
TGT	TGTAA	TGTAT	TGTAC	TGTAG	TGTATA	TGTATT	TGTATC	TGTATG
CAA	CAAAA	CAAT	CAAC	CAAG	CAATA	CAATT	CAATC	CAATG
CAT	CATAA	CATAT	CATAC	CATAG	CATTA	CATT	CATC	CATG
CAC	CACAA	CACAT	CACAC	CACAG	CACATA	CACATT	CACATC	CACATG
CAG	CAGAA	CAGAT	CAGAC	CAGAG	CAGTA	CAGTT	CAGTC	CAGTG
CTA	CTAAA	CTAAT	CTAAC	CTAAG	CTATA	CTATT	CTATC	CTATG
CTT	CTTAA	CTTAT	CTTAC	CTTAG	CTTATA	CTTATT	CTTATC	CTTATG
CTC	CTCAA	CTCAT	CTCAC	CTCAG	CTCTA	CTCTT	CTCTC	CTCTG
CTG	CTGAA	CTGAT	CTGAC	CTGAG	CTGATA	CTGATT	CTGATC	CTGATG
CCA	CCAAA	CCAT	CCAC	CCAG	CCATA	CCATT	CCATC	CCATG
CCT	CCTAA	CCTAT	CCTAC	CCTAG	CCCTA	CCCTT	CCCTC	CCCTG
CCG	CCGAA	CCGAT	CCGAC	CCGAG	CCGTA	CCGTT	CCGTC	CCGTG
CGA	CGAAA	CGAA	CGAC	CGAG	CGATA	CGATT	CGATC	CGATG
CGT	CGTAA	CGTAT	CGTAC	CGTAG	CGTTA	CGTTT	CGTTC	CGTTG
CGC	CGCAA	CGCAT	CGCAC	CGCAG	CGCTA	CGCTT	CGCTC	CGCTG
CGG	CGGAA	CGGAT	CGGAC	CGGAG	CGGTA	CGGTT	CGGTC	CGGTG
GAA	GAAAA	GAAAT	GAAAC	GAAAG	GAAATA	GAAATT	GAAATC	GAAATG
GAT	GATAA	GATAT	GATAC	GATAG	GATATA	GATATT	GATATC	GATATG
GAC	GACAA	GACAT	GACAC	GACAG	GACTA	GACTT	GACTC	GACTG
GAG	GAGAA	GAGAT	GAGAC	GAGAG	GAGTA	GAGTT	GAGTC	GAGTG
GTA	GTAAA	GTAAT	GTAAC	GTAG	GTTA	GTTAT	GTTATC	GTTATG
GTT	GTTAA	GTTAT	GTTAC	GTTAG	GTTATA	GTTATT	GTTATC	GTTATG
GTC	GTCAA	GTCAT	GTCAC	GTCAG	GTTA	GTTT	GTTTC	GTTTG
GTC	GTCAA	GTCAT	GTCAC	GTCAG	GTTA	GTTT	GTTTC	GTTTG
GTA	GTAAT	GTAAT	GTAAC	GTAG	GTTA	GTTT	GTTTC	GTTTG

fastp 1.0.1, at 2025-12-30 16:26:24